

Hand Gesture Recognition using Surface EMG Signals: Exploratory Data Analysis of the GRABMyoPC Dataset

Introduction

Surface electromyography (sEMG) is a technique used to record the electrical activity of muscles during movement. It is widely used in applications such as hand gesture recognition, prosthetic hand control, rehabilitation and human-computer interaction.

Before building any machine learning model, it is important to understand the dataset. Therefore, Exploratory Data Analysis (EDA) was performed on the GRABMyo dataset. The analysis includes studying the dataset structure, participant information, gesture and trial distribution, missing values, duplicate records and EMG signal characteristics. Different statistical summaries and visualizations, such as histograms, boxplots, correlation heatmaps and PCA, were also used to better understand the data which will help in the next stages of the project, including data preprocessing, feature extraction and model development for hand gesture recognition.

Dataset Description

The GRABMyoPC dataset contains high-density surface EMG recordings collected from healthy participants while performing predefined hand gestures.

Dataset characteristics:

Property	Value
Sessions	3
Participants	43
Gesture Classes	17
Trials per Gesture	7
Total Recordings	15,351
Sampling Frequency	2048 Hz
EMG Channels	32

Each recording consists of synchronized EMG signals collected from multiple electrodes placed around the forearm.

Exploratory Data Analysis

The following analyses were performed:

- Dataset structure analysis
- Session distribution
- Participant analysis
- Gender distribution
- Age distribution
- Forearm length distribution
- Upper ring circumference distribution
- Wrist circumference distribution
- Gesture distribution
- Trial distribution
- Missing value analysis
- Duplicate data analysis
- EMG waveform visualization
- Signal histogram
- EMG channel boxplots

- Correlation heatmap
- Principal Component Analysis (PCA)

Key Findings

The EDA revealed several important characteristics:

- The dataset contains three recording sessions and 43 participants.
- All 17 gestures contain 903 recordings, indicating a perfectly balanced dataset.
- Each gesture was repeated seven times, resulting in equal trial distribution.
- The participant ages are primarily between 24 and 29 years, indicating a relatively homogeneous participant group.
- No missing values or duplicate EMG samples were observed.
- EMG signals fluctuate around zero, which is typical of muscle electrical activity.
- Correlation analysis indicates that neighboring EMG channels capture related muscle activation patterns.
- PCA demonstrates that the majority of signal samples share similar characteristics while preserving natural variability.

Previous Work

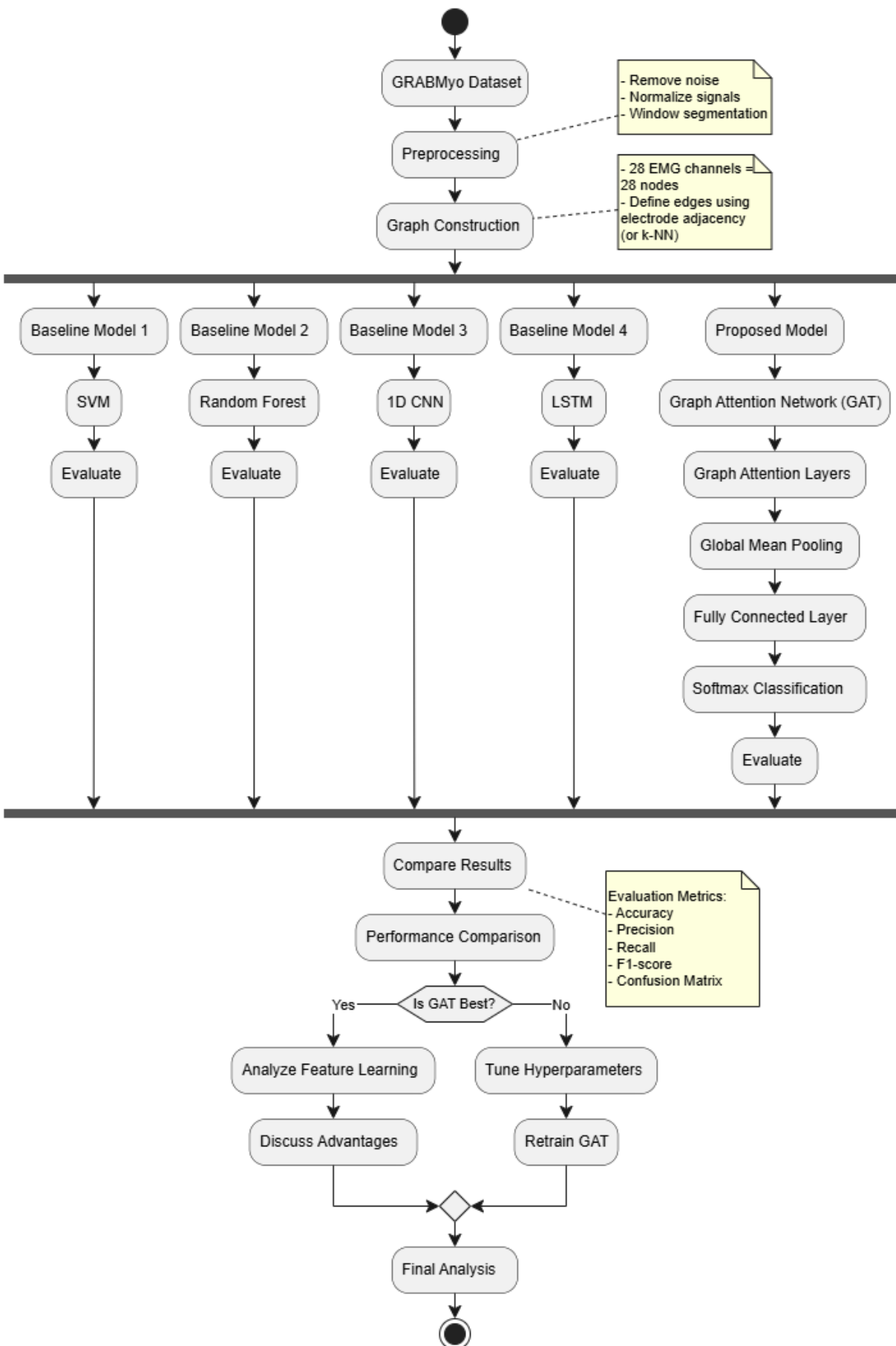
Refer ence	Task / Focus	Model / Archite cture Used	Method	Accura cy / Perfor mance	Researc h Gap	Propos ed Solutio n (Paper)	Our Propose d Work	Limitat ion

Pradhan et al. (2022) [1]	Multi-day sEMG dataset	Dataset only (No ML model)	Longitudinal recording protocol (43 subjects, 3 sessions)	N/A	Lack of large multi-day benchmark datasets	Introduced the GRAB Myo dataset	Use the full GRAB Myo dataset to train and compare multiple deep learning models for gesture recognition.	Dataset quality depends on consistent electrode placement.
Kok et al. (2024) [2]	Prosthetic control	Support Vector Machine (SVM) (compared with KNN & Naïve Bayes)	DWT + composite time-domain feature extraction	SVM outperformed KNN and NB	Hand-crafted features are sensitive to electrode shifts	Feature engineering and feature reduction	Replace manual feature engineering with automatic deep feature learning using CNN-based models.	Requires manual feature extraction.
Bilgin et al. (2023) [3]	Biometric classification	CNN, LSTM	Empirical Mode Decomposition (EMD) + IMF statistical features	CNN: 95.57%, LSTM: 93.88%	Evaluated only 10 subjects performing one gesture	Combined EMD with deep learning	Train on all GRAB Myo gestures and all subjects for more generalized	Small dataset and single-gesture evaluation.

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Aarot ale & Rattan i (2024) [4]	Hand Gesture Recognit ion	1D Dilated CNN	Fused time- domain descripto rs + dilated convoluti ons	97.00%	Session- to- session variabili ty still affects perform ance	Improv ed receptiv e field using dilated CNN	Investiga te more robust architect ures and evaluate generaliz ation across multiple recordin g sessions.	No real- time or hardwa re validati on.
Cansı z et al. (2025) [5]	Gesture + Subject Identific ation	Hierarch ical Cascade Deep Learning Network	Two- stage cascade (gesture → subject)	71.62% Joint Average	Error propagat ion between cascade stages	Separat e gesture recognit ion and biometr ic identific ation	Develop a single end-to- end gesture recogniti on model without cascadin g to reduce accumul ated errors.	Perfor mance depend s on the first- stage predicti on.

Kim et al. (2025) [6]	Biometric Authentication	Convolutional Siamese Network	N-pair loss, shift-equivariant padding, augmentation	EER: 5.62%, 4.92%	Focuses on authentication instead of gesture recognition	Robust cross-day biometric learning	Adapt modern deep learning architectures specifically for multi-class hand gesture recognition rather than authentication.	Limited to biometric verification tasks.
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Proposed Work



Conclusion

The exploratory data analysis confirms that the GRABMyoPC dataset is well organized, balanced and suitable for gesture recognition research as no missing or duplicate signal samples. Also, participant characteristics are consistent and the EMG signals exhibit expected physiological behavior, indicating a strong foundation for developing and evaluating machine learning and deep learning models for hand gesture recognition.

References

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